A FIELD PROJECT REPORT ON

STUDENT ATTENDANCE MANAGEMENT SYSTEM

Submitted

In partial fulfilment of the requirements for the award of the degree

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

Submitted by

K.SRAVANTHI	(241FA04215)
PARITALA TEJA	(241FA04237)
E.HEMA SAI SHREE	(241FA04D47)
O.SAI BHAVAGNA	(241FA04G01)

Under the guidance of

Mr. Kumar Devapogu

Assistant Professor

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SCHOOL OF COMPUTING AND INFORMATICS

VIGNAN'S FOUNDATION FOR SCIENCE, TECHNOLOGY AND RESEARCH (DEEMED TO BE UNIVERSITY) VADLAMUDI, GUNTUR, ANDHRA PRADESH-522213, INDIA

VIGNAN'S
FOUNDATION FOR SCIENCE, TECHNOLOGY & RESEARCH
(Deemed to be University) - Estd. u/s 3 of UGC Act 1956

CERTIFICATE

This is to certify that the field project entitled "STUDENT ATTENDANCE MANAGEMENT SYSTEM" being submitted by (K.SRAVANTHI & 241FA04215), (PARITALA TEJA & 241FA04237), (E.HEMA SAI SHREE & 241FA04D47), and (O.SAI BHAVAGNA& 241FA04G01) in partial fulfilment of Bachelor of Technology in the Department of Computer Science and Engineering, Vignan's Foundation For Science Technology & Research (Deemed to be University), Vadlamudi, Guntur District, Andhra Pradesh, India.

This is a bonafide work carried out by them under my guidance and supervision.

Guide

Project Review Committee **HOD, CSE**

VIGNAN'S
FOUNDATION FOR SCIENCE, TECHNOLOGY & RESEARCH
(Deemed to be University) - Estd. u/s 3 of UGC Act 1956

DECLARATION

We hereby declare that our project work described in the field project titled "**Student Attendance Management System**" is the result of our own efforts and investigations. The project is being submitted under the supervision of **Mr**. **Kumar Devapogu** in partial fulfilment of the requirements for the Bachelor of Technology (B. Tech) degree in the Department of Computer Science and Engineering at Vignan's Foundation For Science, Technology & Research (Deemed to be University), Vadlamudi, Guntur District, Andhra Pradesh, India.

K.SRAVANTHI	(241FA04215)
PARITALA TEJA	(241FA04237)
E.HEMA SAI SHREE	(241FA04D47)
O.SAI BHAVAGNA	(241FA04G01)

Chapter No.		Contents	Page No
1		Introduction	4-5
	1.1	Problem Definition	
	1.2	Existing System	
	1.3	Proposed System	
	1.4	Literature Review	
2		System Requirements	6-7
	2.1	Hardware & Software Requirements	
	2.2	Software Requirements Specification(SRS)	
3		System Design	
	3.1	Modules of System	8-9
	3.2	UML Diagrams	
4		Implementation	10-21
	4.1	Sample Code	
	4.2	Test Cases	
5		Results	22-27
	5.1	Output Screens	
6		Conclusion	28
		References	29

CHAPTER-1

INTRODUCTION

Attendance management is a core academic activity that affects student evaluation and institutional accountability. Traditional manual and semi-digital methods are time-consuming, error-prone, and unsuitable for large or online classes. The Student Attendance Management System is a web-based solution designed to manage attendance in both offline and online modes using role-based access, session-based marking, and automatic report generation, ensuring accuracy and efficiency.

1.1 Problem definition

In many schools and colleges, student attendance is still recorded manually using paper registers. This traditional method takes a lot of time and requires extra effort from teachers. It is also difficult to manage attendance records for a large number of students. Manual attendance systems are prone to errors such as wrong entries, missing data, and miscalculations of attendance percentage.

1.2 Existing system

- Attendance recorded manually using paper registers
- Records stored in files or notebooks
- Time-consuming for large classes
- Prone to human errors (wrong entries, missing data, miscalculations)
- Attendance percentage calculated manually
- Difficult to search and retrieve old records
- Paper records may be lost or damaged
- Report generation is slow and inefficient

1.3 Proposed System

- Computer-based Student Attendance Management System
- Attendance recorded digitally instead of paper registers
- Teachers can mark students as **Present** or **Absent** easily
- Attendance data stored securely in digital format
- Reduces manual work and paperwork
- Automatically calculates attendance percentage
- Attendance records can be accessed anytime when required

1.4 Literature Review

Numerous studies and projects have focused on student attendance management systems. Traditionally, educational institutions relied on manual methods, such as paper registers, to record attendance. These approaches were not only time-consuming but also prone to errors, including incorrect entries and inaccurate attendance calculations. Subsequent research introduced computerized attendance systems, enabling institutions to maintain digital records efficiently. Evidence indicates that these digital systems significantly enhance accuracy, streamline attendance tracking, and reduce the administrative burden on teachers.

CHAPTER-2

SYSTEM REQUIREMENTS

2.1 Hardware and software requirements

Hardware Requirements

1. Computer/Server
    - Processor: Intel i3 or higher
    - RAM: 4 GB minimum (8 GB recommended for larger institutions)
    - Storage: 500 GB HDD/SSD (depends on data volume)
    - Monitor: 15-inch or larger
2. Input Devices
    - Keyboard and Mouse
3. Network
    - LAN connection or Wi-Fi for centralized systems
    - Internet connection (optional, for cloud-based or online systems)

Software Requirements

1. Operating System
    - Windows 10/11 or Linux (Ubuntu)
2. Development Tools
    - IDEs: Visual Studio
3. Web/Front-End (if web-based)
    - Front-end: HTML, CSS, JavaScript

2.2 Software requirements specifications (SRS)

Functional Requirements

1. User Authentication:
    - Login for teachers, students, and admin with unique credentials.
2. Attendance Management:
    - Mark daily attendance for each student.
    - Support multiple classes, sections, and subjects.
3. Report Generation:
    - Generate attendance summary by student number.

Non Functional Requirements:

- Performance: Fast and responsive system for quick attendance marking.
- Usability: Easy to use with a user-friendly interface for faculty and students.
- Security: High data protection for student records and secure login credentials.

- Availability: 24/7 system access for viewing reports and summaries.
- Integrity: Ensures accurate, tamper-proof attendance data once submitted.
- Scalability: Capable of handling multiple classes, sections, and subjects.

CHAPTER 3

SYSTEM DESIGNS

3.1 Modules of System

1. User Module:

- Login to the system
- Add student details
- View student list
- Mark students as Present or Absent
- View attendance records
- Generate attendance reports

2. Admin Module:

- Manage teacher accounts
- Add, update, or delete student details
- View overall attendance data
- Generate summary reports
- Monitor system usage

3. Student Module:

- View personal attendance details
- Check attendance percentage
- View attendance reports

4. Attendance Management Module:

- Record daily attendance
- Update attendance records
- Calculate attendance percentage automatically
- Store attendance data securely

5. Report Module:

- Generate daily attendance reports
- Generate monthly attendance reports
- Display attendance percentage

- Download or print reports

6. Data Storage Module:

- Store student and attendance data digitally
- Retrieve data when required
- Maintain accuracy of records

3.2 UML Diagrams:

1. Use Case Diagram:

- Displays interactions between the Teacher (marking attendance), Student (viewing reports), and Admin (managing credentials)

2. Class Diagram:

- Defines the structure of classes like Student, Teacher, AttendanceRecord, and Subject, and how they relate.

3.Activity Diagram:

- Illustrates the operational flow, such as a teacher selecting a class, marking students present/absent, and saving the record.

4.Sequence Diagram:

- Shows the time-ordered interaction where a user requests an attendance summary and the system retrieves it from the database

CHAPTER 4

IMPLEMENTATION

Source Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Student Attendance Management System</title>
<style>
body{
font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
background: linear-gradient(135deg, #e0eafc, #cfdef3);
margin:0;
min-height: 100vh;
}
h1{
background: #1e3a8a; /* Deep Blue */
color:white;
padding:20px;
text-align:center;
margin: 0;
box-shadow: 0 4px 6px rgba(0,0,0,0.1);
}
.container{
width:90%;
```

```
max-width: 800px;
margin:30px auto;
background:white;
padding:25px;
border-radius:12px;
box-shadow: 0 10px 25px rgba(0,0,0,0.1);
}
h2 {
color: #1e3a8a;
border-bottom: 2px solid #3b82f6;
padding-bottom: 10px;
}
select,input,button{
padding:10px;
margin:8px 5px;
border-radius: 6px;
border: 1px solid #cbd5e1;
}
button{
background: #10b981; /* Emerald Green */
color:white;
border:none;
cursor:pointer;
font-weight: bold;
transition: background 0.3s ease;
}
button:hover{
```

```
background:#059669;
}
table{
width:100%;
border-collapse:collapse;
margin-top:20px;
background: #f8fafc;
}
th,td{
border:1px solid #e2e8f0;
padding:12px;
text-align:center;
}
th{
background: #3b82f6; /* Bright Blue */
color:white;
text-transform: uppercase;
font-size: 0.9em;
}
tr:nth-child(even) {
background-color: #f1f5f9;
}
.section{
margin-top:30px;
}
#result {
margin-top: 20px;
```

```
padding: 15px;
background: #eff6ff;
border-radius: 8px;
border-left: 5px solid #3b82f6;
}
.present{color:#059669;font-weight:bold;}
.absent{color:#dc2626;font-weight:bold;}
.hidden{display:none;}
input[type="checkbox"] {
transform: scale(1.5);
cursor: pointer;
}
</style>
</head>
<body>
<h1>Student Attendance Management System</h1>
<div class="container" id="loginPanel">
<h2>Login</h2>
<select id="roleSelect">
<option value="">Select Role</option>
<option value="teacher">Teacher</option>
<option value="student">Student</option>
</select><br>
<input type="text" id="username" placeholder="Username"><br>
<input type="password" id="password" placeholder="Password"><br>
<button onclick="login()">Login to Portal</button>
</div>
```

```
<div class="container hidden" id="teacherPanel">
<h2>Teacher Panel - Mark Attendance</h2>
<select id="dept">
<option value="">Select Department</option>
<option value="CSE">CSE</option>
<option value="ECE">ECE</option>
<option value="EEE">EEE</option>
</select>
<select id="section">
<option value="">Select Section</option>
<option value="A">Section A (1)</option>
<option value="B">Section B (2)</option>
<option value="C">Section C (3)</option>
</select>
<button onclick="loadStudents()" style="background:#3b82f6;">Load Class List</button>
<table>
<thead>
<tr>
<th>Roll No</th>
<th>Name</th>
<th>Department</th>
<th>Section</th>
<th>Tick if Absent</th>
</tr>
</thead>
<tbody id="attendanceTable"></tbody>
</table>
```

```
<button onclick="saveAttendance()" style="width: 100%; margin-top: 20px;">Submit
Attendance</button>
</div>
<div class="container section hidden" id="studentPanel">
<h2>Student Panel - View Attendance</h2>
<select id="studentSelect">
<option value="">Select Student</option>
</select>
<button onclick="viewAttendance()">View My Report</button>
<div id="result"></div>
</div>
<script>
const users = [
{username:"teacher1", password:"teach123", role:"teacher"},
{username:"student1", password:"stud101", roll:101, role:"student"},
{username:"student2", password:"stud102", roll:102, role:"student"},
{username:"student3", password:"stud103", roll:103, role:"student"},
{username:"student4", password:"stud104", roll:104, role:"student"},
{username:"student5", password:"stud105", roll:105, role:"student"},
{username:"student6", password:"stud201", roll:201, role:"student"},
{username:"student7", password:"stud202", roll:202, role:"student"},
{username:"student8", password:"stud203", roll:203, role:"student"},
{username:"student9", password:"stud204", roll:204, role:"student"},
{username:"student10", password:"stud205", roll:205, role:"student"},
{username:"student11", password:"stud301", roll:301, role:"student"},
{username:"student12", password:"stud302", roll:302, role:"student"},
{username:"student13", password:"stud303", roll:303, role:"student"},
```

```
{username:"student14", password:"stud304", roll:304, role:"student"},
{username:"student15", password:"stud305", roll:305, role:"student"}
];
const sections = {
A:{number:1,name:"Section A"},
B:{number:2,name:"Section B"},
C:{number:3,name:"Section C"}
};
const students = [
{roll:101,name:"Ravi",dept:"CSE",sec:"A"},
{roll:104,name:"Arjun",dept:"CSE",sec:"A"},
{roll:105,name:"Nikhil",dept:"CSE",sec:"A"},
{roll:102,name:"Suresh",dept:"CSE",sec:"B"},
{roll:106,name:"Varun",dept:"CSE",sec:"B"},
{roll:107,name:"Ajay",dept:"CSE",sec:"B"},
{roll:103,name:"Kiran",dept:"CSE",sec:"C"},
{roll:108,name:"Prakash",dept:"CSE",sec:"C"},
{roll:109,name:"Deepak",dept:"CSE",sec:"C"},
{roll:201,name:"Anu",dept:"ECE",sec:"A"},
{roll:204,name:"Kavya",dept:"ECE",sec:"A"},
{roll:205,name:"Sneha",dept:"ECE",sec:"A"},
{roll:202,name:"Divya",dept:"ECE",sec:"B"},
{roll:206,name:"Pooja",dept:"ECE",sec:"B"},
{roll:207,name:"Ritika",dept:"ECE",sec:"B"},
{roll:203,name:"Meena",dept:"ECE",sec:"C"},
{roll:208,name:"Nandini",dept:"ECE",sec:"C"},
{roll:209,name:"Shreya",dept:"ECE",sec:"C"},
```

```
{roll:301,name:"Manoj",dept:"EEE",sec:"A"},
{roll:304,name:"Karthik",dept:"EEE",sec:"A"},
{roll:305,name:"Surya",dept:"EEE",sec:"A"},
{roll:302,name:"Rohit",dept:"EEE",sec:"B"},
{roll:306,name:"Vikram",dept:"EEE",sec:"B"},
{roll:307,name:"Sathish",dept:"EEE",sec:"B"},
{roll:303,name:"Sanjay",dept:"EEE",sec:"C"},
{roll:308,name:"Harish",dept:"EEE",sec:"C"},
{roll:309,name:"Lokesh",dept:"EEE",sec:"C"}
];
let attendance = JSON.parse(localStorage.getItem("attendance")) || {};
let currentUser = null;
function login(){
const role = roleSelect.value;
const user = users.find(
u => u.username===username.value &&
u.password===password.value &&
u.role===role
);
if(!user){ alert("Invalid Login"); return; }
currentUser = user;
loginPanel.classList.add("hidden");
if(role=="teacher"){
teacherPanel.classList.remove("hidden");
studentPanel.classList.remove("hidden");
}else{
studentPanel.classList.remove("hidden");
```

```
studentSelect.innerHTML = `<option value="${user.roll}">${user.roll}</option>`;
}
}
function loadStudents(){
attendanceTable.innerHTML="";
studentSelect.innerHTML='<option value="">Select Student</option>';
students.filter(s=>s.dept===dept.value && s.sec===section.value)
.forEach(s=>{
attendanceTable.innerHTML+=`
<tr>
<td>${s.roll}</td>
<td>${s.name}</td>
<td>${s.dept}</td>
<td>${sections[s.sec].name}</td>
<td><input type="checkbox" id="absent_${s.roll}"></td>
</tr>`;
studentSelect.innerHTML+=`<option value="${s.roll}">${s.name}</option>`;
});
}
function saveAttendance(){
students.forEach(s=>{
const cb=document.getElementById("absent_"+s.roll);
if(cb){
if(!attendance[s.roll]) attendance[s.roll]={info:s,present:0,total:0};
attendance[s.roll].total++;
if(!cb.checked) attendance[s.roll].present++;
}
```

```
});
localStorage.setItem("attendance",JSON.stringify(attendance));
alert("Attendance Saved Successfully!");
}
function viewAttendance(){
const roll=currentUser.role=="student"?currentUser.roll:Number(studentSelect.value);
if(!attendance[roll]){ result.innerHTML="No record found for this student."; return; }
const d=attendance[roll];
result.innerHTML=`
<h3>Attendance Summary: ${d.info.name}</h3>
<p>Total Classes Held: <strong>${d.total}</strong></p>
<p class="present">Total Present: ${d.present}</p>
<p class="absent">Total Absent: ${d.total-d.present}</p>
<p>Attendance Percentage: <strong>${((d.present/d.total)*100).toFixed(2)}%</strong></p>`;
}
</script>
</body>
</html>
```

Explanation:

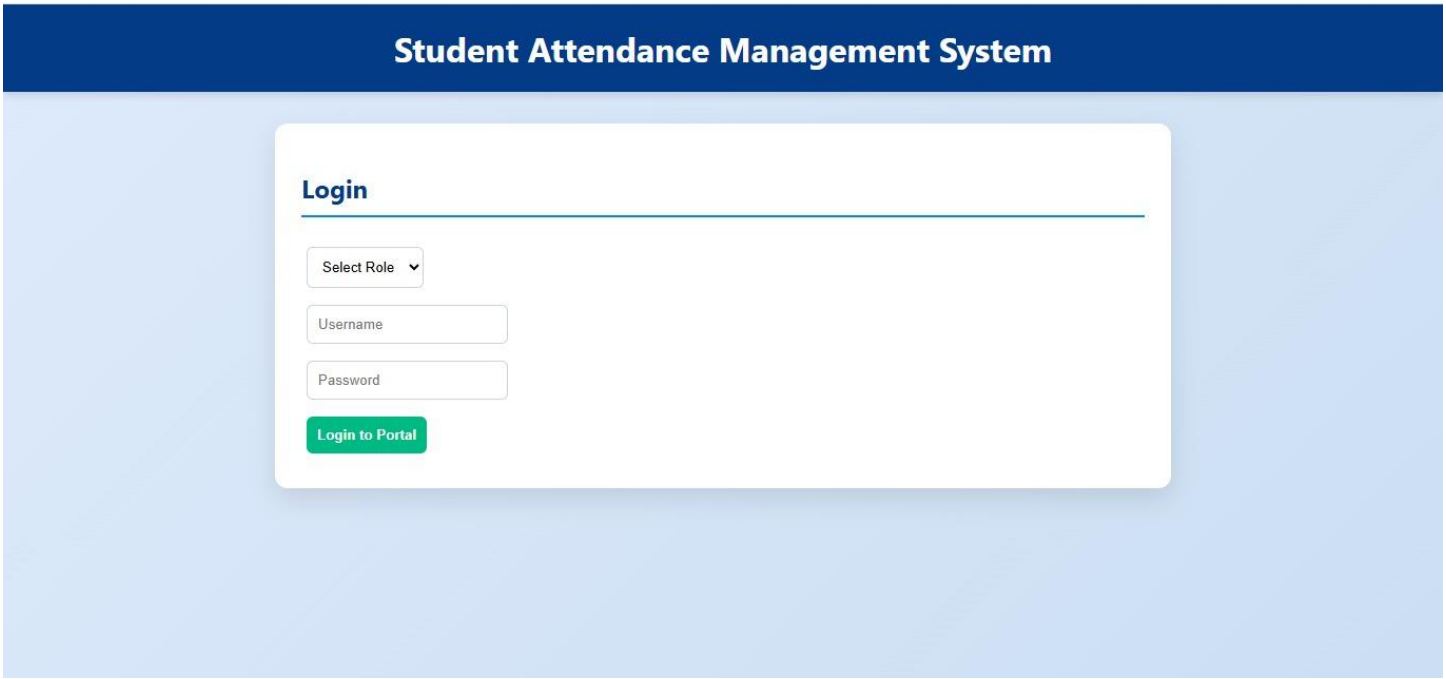
- Declares HTML5 document and sets language to English.
- Includes internal css and javascript concepts
- Contains div tags
- Shows a pop-up for invalid login credentials and successful marking of attendance
- Filtering: The loadStudents() function uses the .filter() method to match students to the selected department and section values.
- Marking: The system assumes all students are "Present" by default. If the "Tick if Absent" checkbox is selected, the system logs that student as absent.
- Calculation: The system calculates the percentage using the formula: (Present / Total) x 100

CHAPTER 5

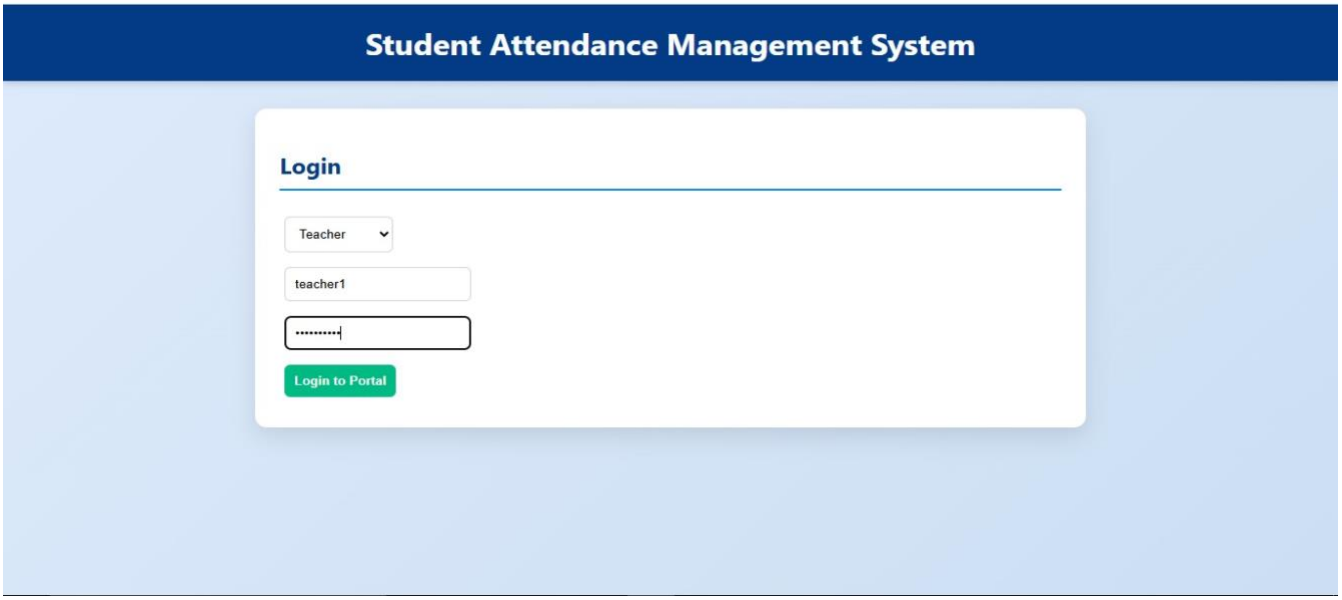
RESULTS

5.1 Output Screens

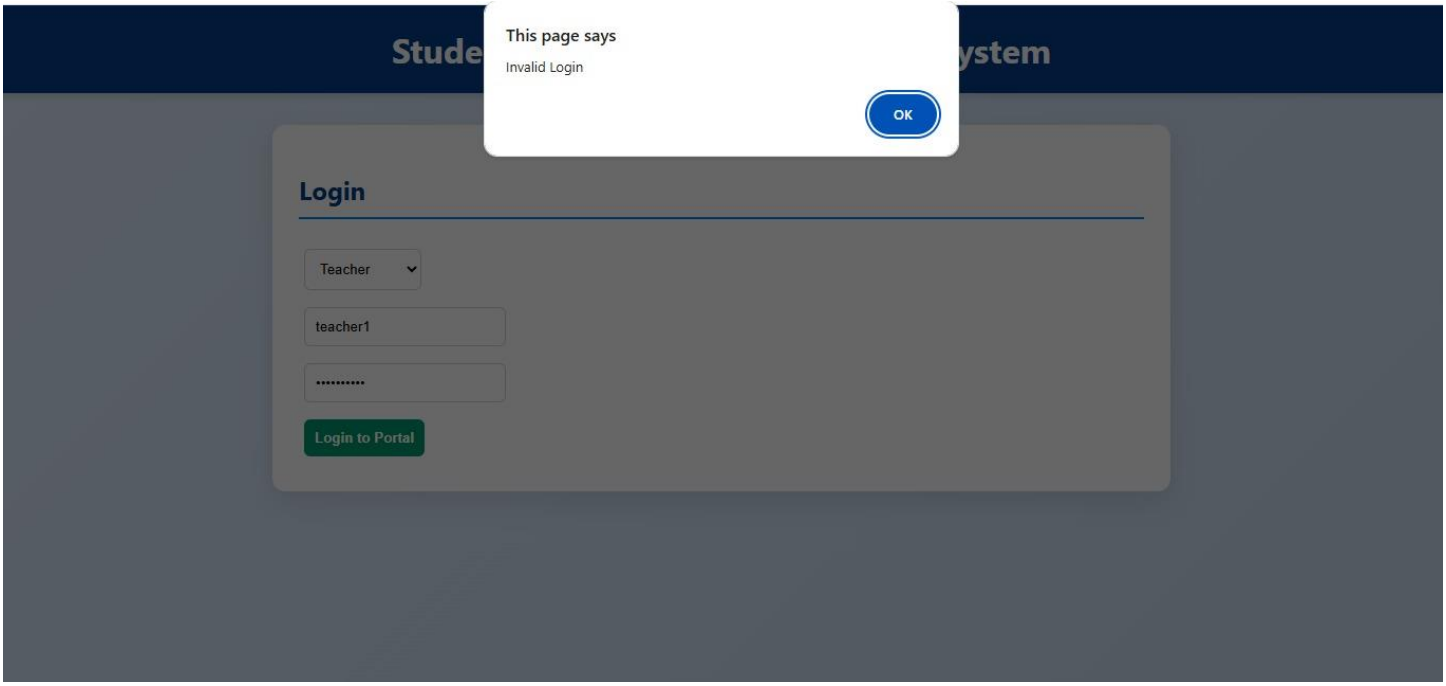
Main Page:



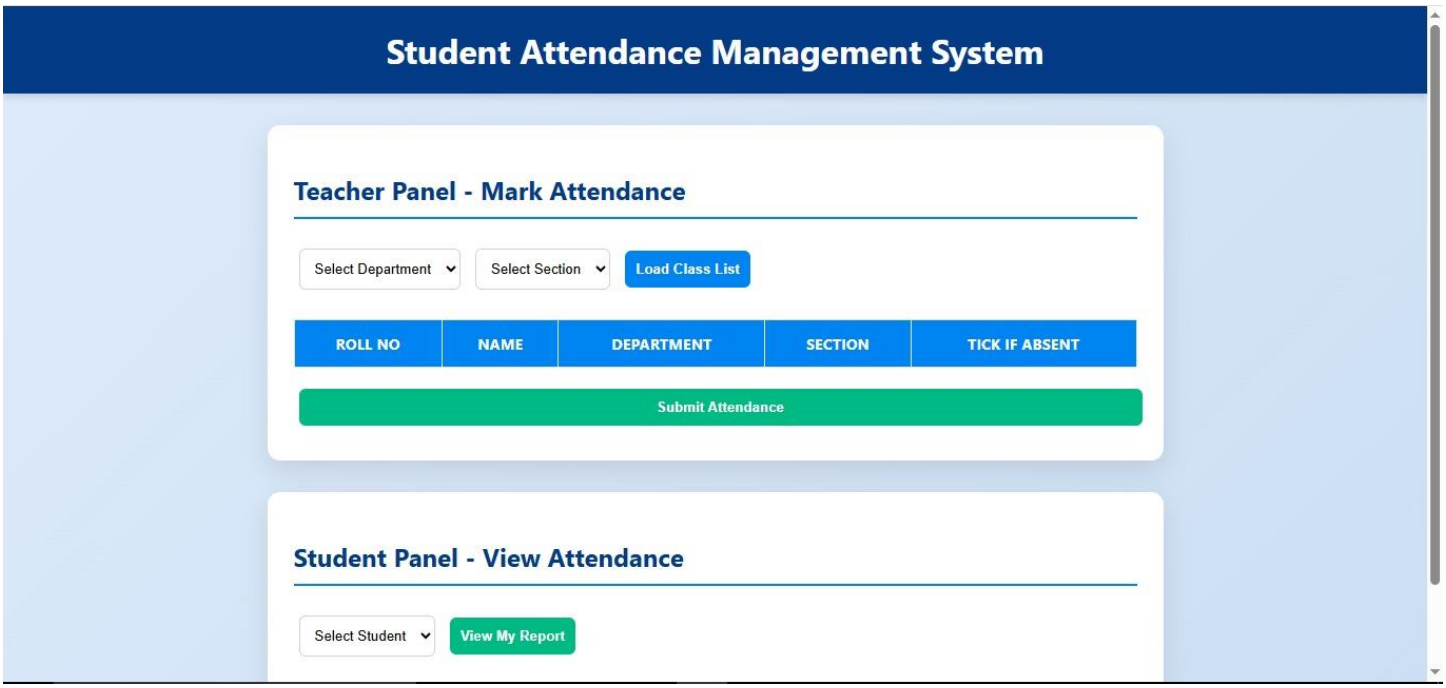
Teacher Login:



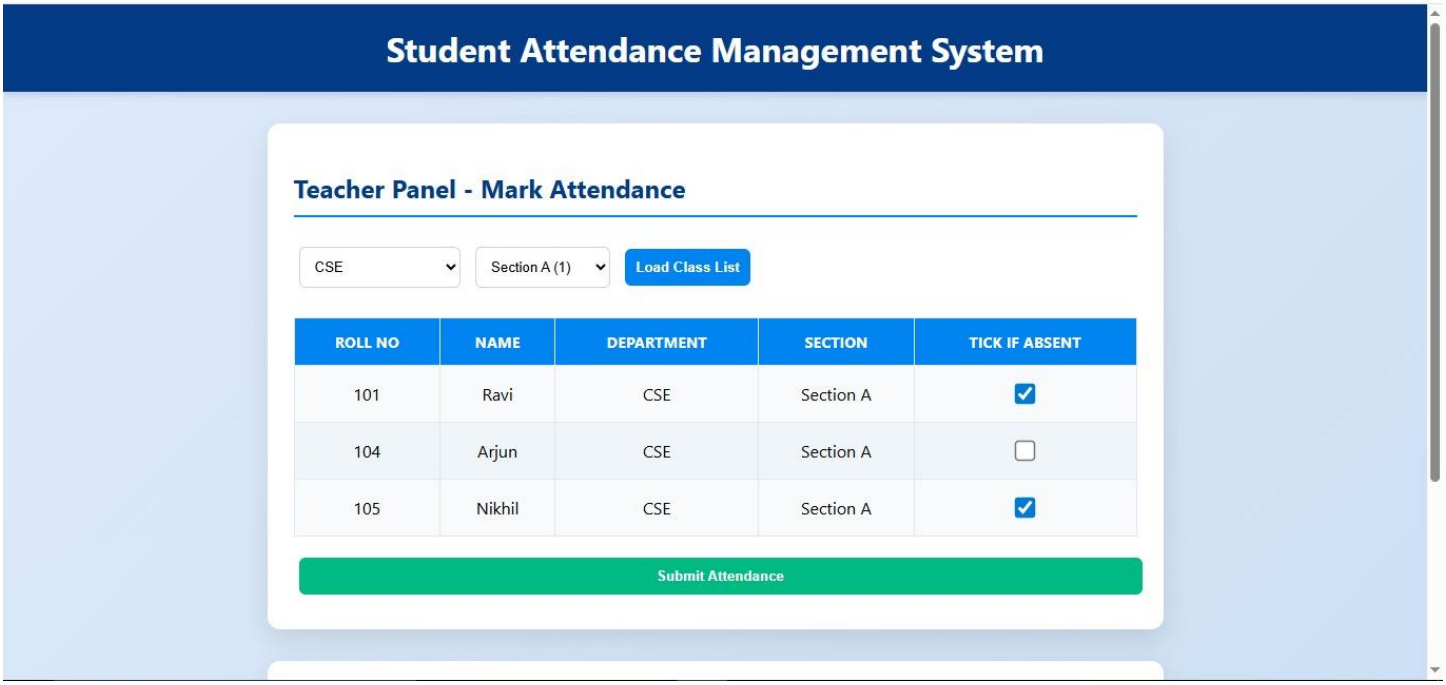
Invalid Login:



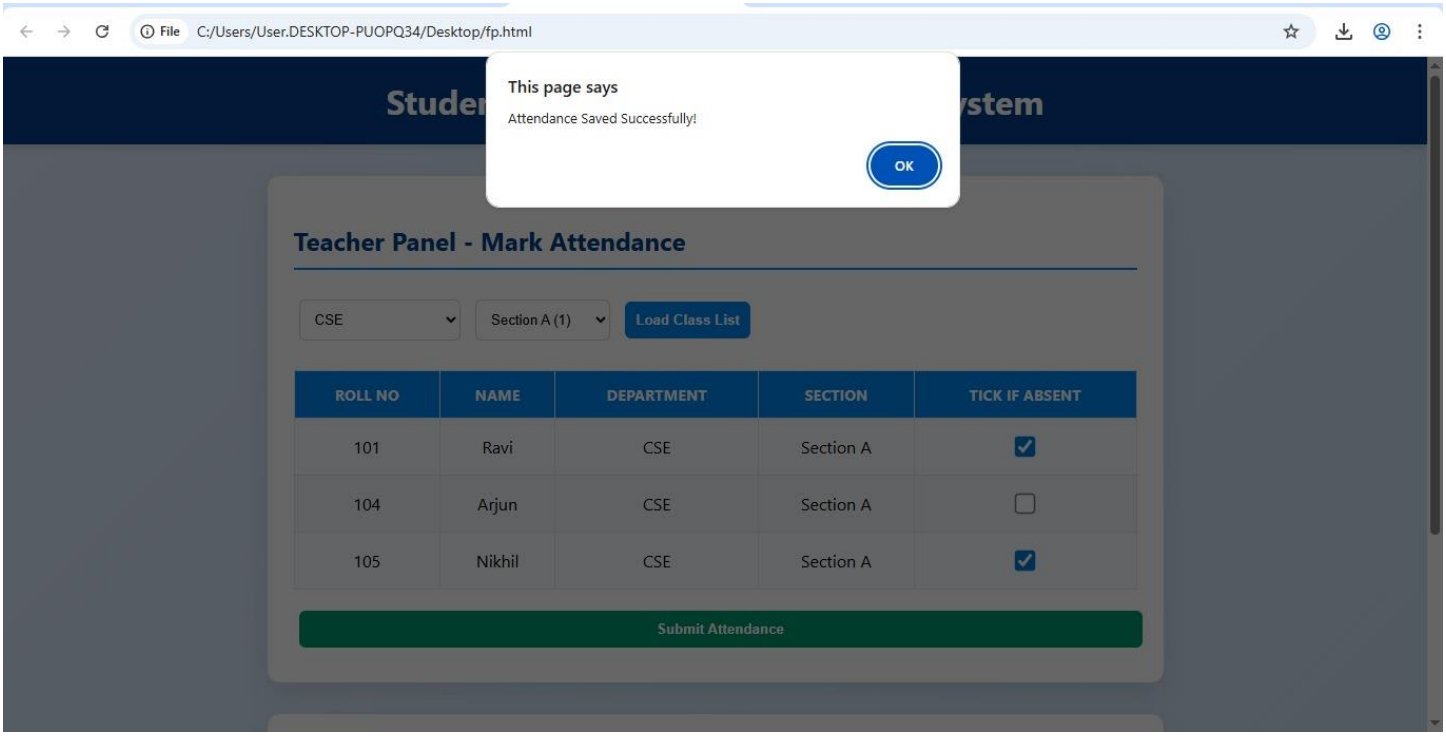
Pre-view of the portal before selecting:



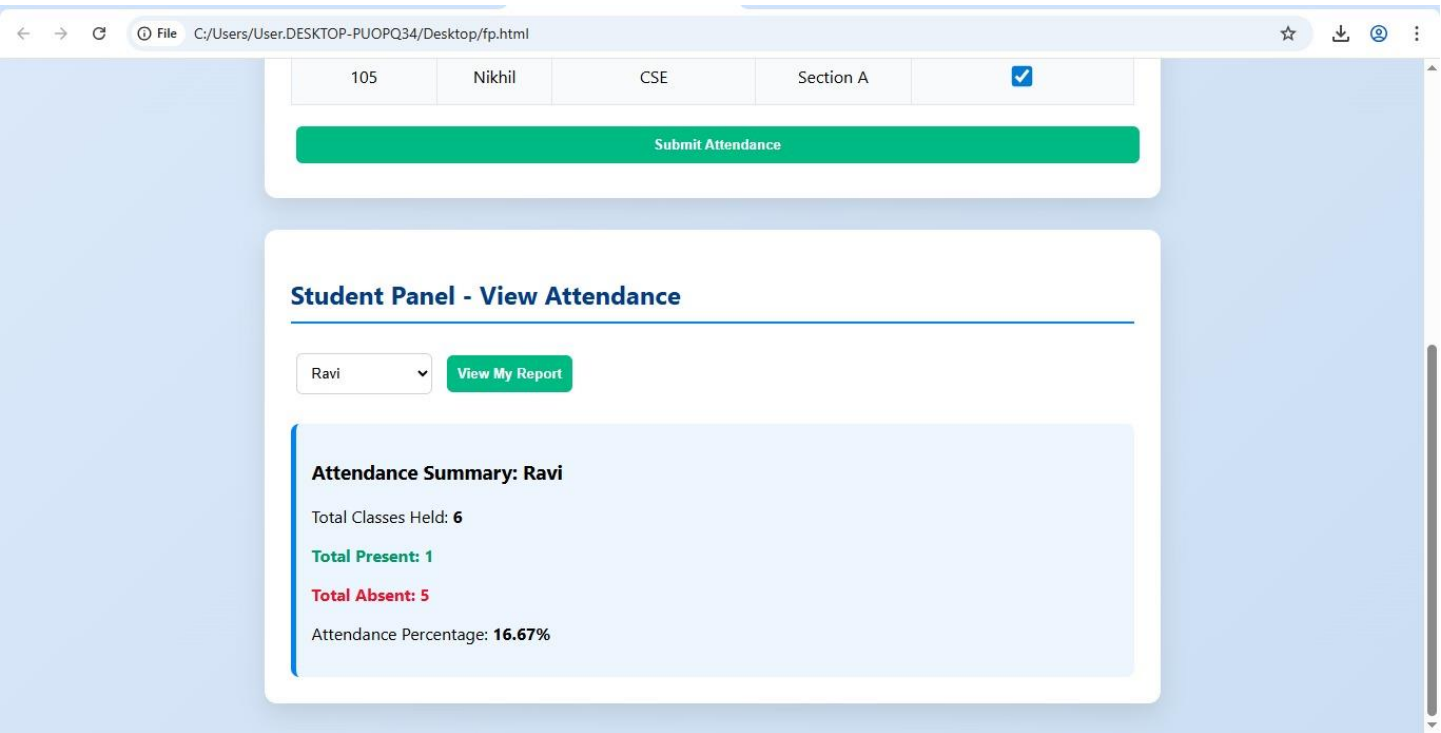
Teacher Panel:



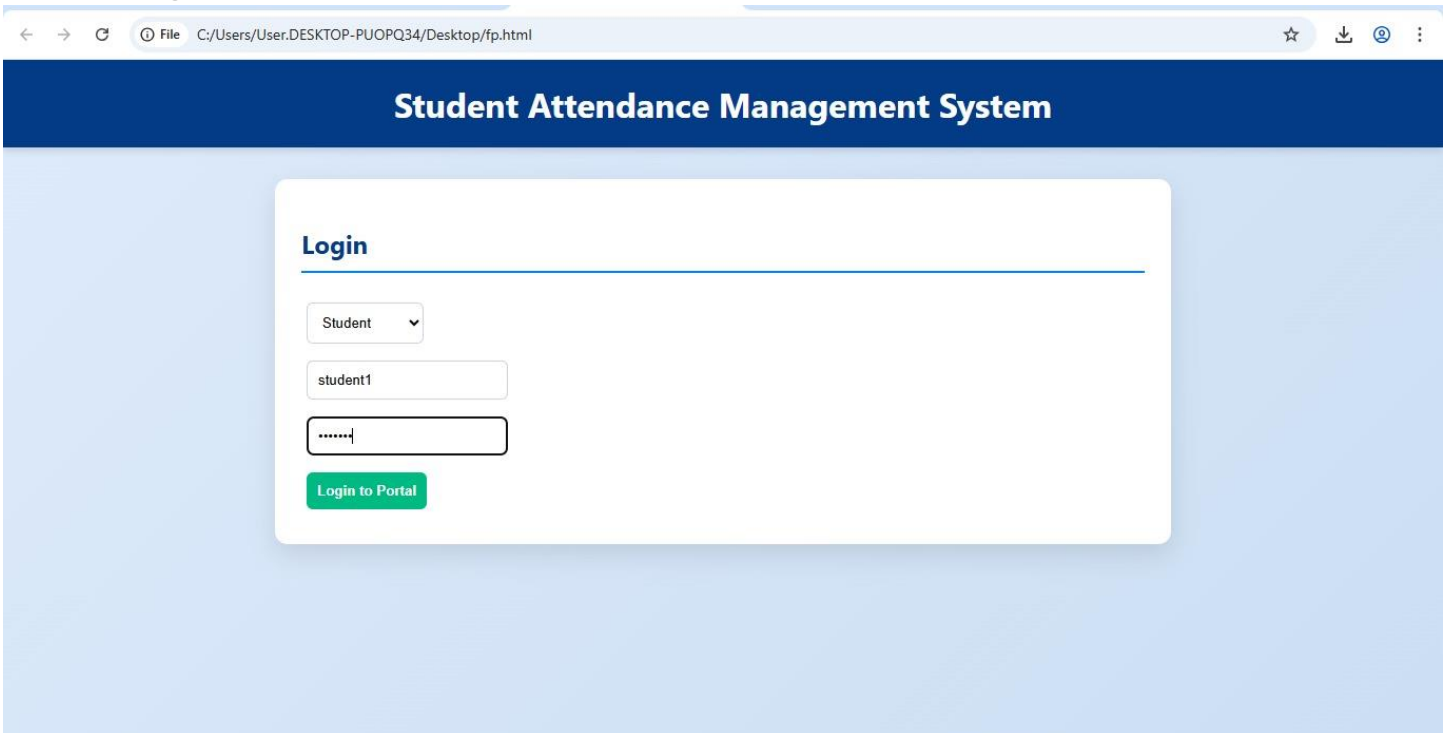
Pop-up after attendance:



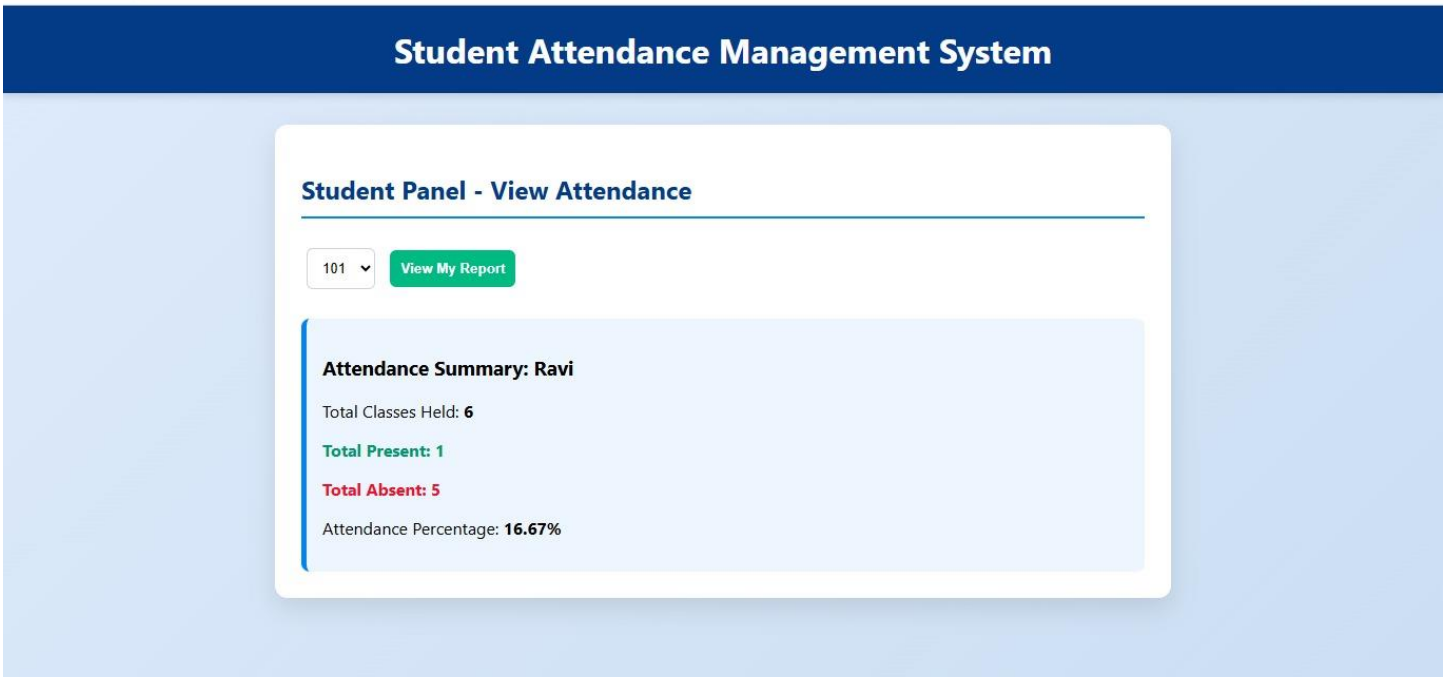
Teacher panel to view student attendance:



Student login:



Student panel to view attendance:



CHAPTER 6

CONCLUSION

The Student Attendance Management System provides an efficient alternative to traditional paper-based methods. By utilizing modern web technologies, it ensures data integrity, ease of access, and streamlined operations for both students and faculty.

CHAPTER 7

REFERENCES

7.1 Academic Journals & Papers

1. Chintalapati, S., & Raghunadh, M. V. (2013). "Automated Attendance Management System based on Face Recognition Algorithms." *International Conference on Control, Automation, Communication and Energy Conservation.*

2. Jain, A., & Singh, R. (2018). "Web-Based Student Attendance Management System." *International Journal of Computer Applications.* 3. Shoewu, O., & Idowu, O. A. (2012). "Development of Attendance Management System using Biometrics." *Pacific Journal of Science and Technology.*

7.2 Technical Documentation

4. MDN Web Docs. "JavaScript: The building blocks of the Web." https://developer.mozilla.org

5. W3Schools. "HTML5 and CSS3 Responsive Web Design." https://www.w3schools.com

6. Microsoft Documentation. "Visual Studio Code and Web Development." https://code.visualstudio.com/docs